

50 Important Java Interview Questions for Freshers

This PDF contains commonly asked Java interview questions and short answers useful for fresher interviews, job melas, and technical rounds.

Q1. What is Java?

Answer: Java is a high-level, object-oriented, platform-independent programming language.

Q2. What are the features of Java?

Answer: Platform independence, OOP, robustness, security, multithreading, portability.

Q3. What is JVM?

Answer: Java Virtual Machine executes Java bytecode.

Q4. What is JDK?

Answer: Java Development Kit contains JRE and development tools.

Q5. What is JRE?

Answer: Java Runtime Environment provides libraries and JVM to run Java programs.

Q6. Difference between JDK, JRE, JVM?

Answer: JDK = development kit, JRE = runtime environment, JVM = executes bytecode.

Q7. What is OOP?

Answer: Object-Oriented Programming organizes programs using objects and classes.

Q8. What are the four pillars of OOP?

Answer: Encapsulation, Inheritance, Polymorphism, Abstraction.

Q9. What is a class?

Answer: A blueprint for creating objects.

Q10. What is an object?

Answer: An instance of a class.

Q11. What is inheritance?

Answer: Mechanism where one class acquires properties of another.

Q12. What is polymorphism?

Answer: Ability of a method to behave differently.

Q13. What is abstraction?

Answer: Hiding implementation details and showing functionality.

Q14. What is encapsulation?

Answer: Binding data and methods together.

Q15. Difference between overloading and overriding?

Answer: Overloading occurs in same class with different parameters; overriding changes parent method behavior.

Q16. What is constructor?

Answer: Special method used to initialize objects.

Q17. Types of constructors?

Answer: Default and parameterized constructors.

Q18. What is static keyword?

Answer: Belongs to class rather than object.

Q19. What is final keyword?

Answer: Used to restrict modification, inheritance, or overriding.

Q20. What is this keyword?

Answer: Refers to current object.

Q21. What is super keyword?

Answer: Refers to parent class object.

Q22. What is an interface?

Answer: A blueprint containing abstract methods.

Q23. What is an abstract class?

Answer: Class that cannot be instantiated and may contain abstract methods.

Q24. Difference between abstract class and interface?

Answer: Abstract class can have concrete methods; interface mainly contains abstract methods.

Q25. What is method overriding?

Answer: Redefining parent class method in child class.

Q26. What is method overloading?

Answer: Same method name with different parameters.

Q27. What is exception handling?

Answer: Mechanism to handle runtime errors.

Q28. Keywords used in exception handling?

Answer: try, catch, finally, throw, throws.

Q29. Difference between checked and unchecked exceptions?

Answer: Checked exceptions are checked at compile time; unchecked at runtime.

Q30. What is multithreading?

Answer: Executing multiple threads simultaneously.

Q31. What is thread?

Answer: Smallest unit of execution.

Q32. Difference between process and thread?

Answer: Process has separate memory; threads share memory.

Q33. What is synchronization?

Answer: Controls access to shared resources in multithreading.

Q34. What is String in Java?

Answer: Immutable sequence of characters.

Q35. Difference between String and StringBuilder?

Answer: String is immutable; StringBuilder is mutable.

Q36. What is ArrayList?

Answer: Resizable array implementation in Java Collections.

Q37. Difference between Array and ArrayList?

Answer: Array has fixed size; ArrayList is dynamic.

Q38. What is Collection Framework?

Answer: Architecture for storing and manipulating groups of objects.

Q39. What is HashMap?

Answer: Stores key-value pairs.

Q40. Difference between HashMap and Hashtable?

Answer: HashMap is not synchronized; Hashtable is synchronized.

Q41. What is package in Java?

Answer: Group of related classes and interfaces.

Q42. What is access modifier?

Answer: Controls accessibility of classes and methods.

Q43. Types of access modifiers?

Answer: public, private, protected, default.

Q44. What is garbage collection?

Answer: Automatic memory management process.

Q45. What is recursion?

Answer: Method calling itself repeatedly.

Q46. What is wrapper class?

Answer: Converts primitive data types into objects.

Q47. What is autoboxing?

Answer: Automatic conversion of primitive to wrapper object.

Q48. What is type casting?

Answer: Converting one datatype into another.

Q49. What is Java API?

Answer: Predefined libraries and classes provided by Java.

Q50. Why is Java platform independent?

Answer: Because Java code runs on JVM regardless of OS.
